Mth401 Short Notes Final Term



Scanned with CamScanner

MTH 501
important short + long
Day: M T W T F S Date: / /20
Rashid let 25 to 30

(1) Let W be the subspace of set of all (2×2) matrices defined by

$$W = \{A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : 2a - b + 3c + d = 0\}$$

(2) Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + d \\ b - 2c - d \end{bmatrix} : a, b, c, d \in R \right\}$$

(3) Find a basis for Col B, where B

$$B = [b_1 \; b_2 \; \cdots \; b_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(4) Let $v_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} -2 \\ 7 \\ -9 \end{bmatrix}$. Determine if $\{v_1, v_2\}$ is a basis for R^3. is $[v_1, v_2]$ a basis for R^2

(5) the matrices $A = \begin{bmatrix} 2 & -1 & 1 & -6 & 8 \\ 1 & -2 & -4 & 3 & -2 \\ -7 & 8 & 10 & 3 & -10 \\ 4 & -5 & -7 & 0 & 4 \end{bmatrix}$

Rashid Hussain

$$B = \begin{bmatrix} 1 & -2 & -4 & 3 & -2 \\ 0 & 3 & 9 & -12 & 12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(i) Find Rank A and dim Null A

(ii) Find bases for Col A and Row A

(iii) How many pivot columns are in Row echelon form (Ans) $Col A^t = Row$, so A^t two pivot Colu

Scanned with CamScanner

Day: M T W T F S S Date: __/__/20__

(6) $b_1 = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$, $c_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $c_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$

Consider the bases given $B = \{b_1, b_2\}$ and $C = \{c_1, c_2\}$. Find the change of coordinates matrix from B to C

Rashid Hussain

(7) verify 1^k, $(-4)^k$ and 3^k are linearly independent signals.

(8) The Casoratic matrix $K = 0$ is invertible so, 1^k, $(-4)^k$, 3^k are linearly indepen.

(9) $y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k + 6y_k = 0$

$y_k = z^k$ for some z. Find some solution of the eq.

(9) write the following difference equation as a first order System.

$y_{k+3} - 2y_{k+1} + 6y_k = 0$ for all k

(10) let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ are u and v eigenvector of A?

lect 29,30 Ans $Au = \lambda[u]$ (hint)

(multiple of u)

(11) the characteristic polynomial of a 6x6 matrix is $\lambda^6 - 4\lambda^5 - 12\lambda^4$ find the eigenvalues and their multiplicities.

Scanned with CamScanner

MTH401
Lect #40
short questions for final term

Rashid Hussain

Q.No.1 write the following non-homogeneous system of differential equations in the matrix form

$$\frac{dx}{dt} = -2x + 5y + e^t - 2t$$
$$\frac{dy}{dt} = 4x - 3y + 10t$$

Solution: if we suppose that

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

Then, the given non-homogeneous diff eq. we can be written as

$$\frac{dx}{dt} = \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} e^t - 2t \\ 10t \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} -2 \\ 10 \end{pmatrix} t$$

(Short)

Q.No.2 verify that

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$$

are solution of the following system of the homogeneous diff. eq.

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

on the interval $(-\infty, \infty)$

Solution:

Since $X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$

$$X_1' = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$$

Scanned with CamScanner

Day: M T W T F S Date: __/__/20

(19) Apply the power method to $A = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}$

$x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. stop when $k = 5$, and estimate the dominant eigenvalue and a corresponding eigenvector of A.

(Hints) $Ax_0 = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $\mu_0 = 5$

$x_1 = \frac{1}{\mu_0} Ax_0$ to get x_1

$x_2 = \frac{1}{\mu_1} Ax_1$ to get x_2

$x_3 = \frac{1}{\mu_2} Ax_2$ to get x_3

(20) $x_{k+1} = \left(\frac{1}{\mu_k}\right) Ax_k$

Rashid Hussain

Today Paper

1. Page 311 Example 4
2. Page 371 example 6 (related)
3. Page 411 exapla 4
4. Page 460 Example 1
5. Page 494 Exapla 1 (related)

Scanned with CamScanner

Day: (illegible) Date: __/__/20

(17) Define $T: \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x) = Ax$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a basis B for $\mathbb{R}^2$ with the property that the B-matrix of T is a diagonal matrix.

Ans Key Point first = eigen values

2nd = Eigen vector

$P = \begin{bmatrix} \text{Eigen vector} \\ \text{values} \end{bmatrix}$ 3rd = $A = PDP^{-1}$

$D = \begin{bmatrix} \text{Eigen} \\ \text{values} \end{bmatrix}$ $\therefore A = PDP^{-1}$

(18) $P = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ $P^{-1}AP = D$

$D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$ $D = B$

$B = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$ $\boxed{P^{-1}AP = B}$

(18) Show that the origin is a saddle point of solution of $x_{k+1} = Ax_k$, where

$$A = \begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$$

Find the directions of greatest attraction and greatest repulsion.

Ans hints: First find eigen values $\lambda_1 = 2$ $\lambda_2 = .5$

2nd eigen vector ~~$\lambda_1 = 2$~~

$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. $|2| > 1$, $|.5| < 1$

the origin is a saddle point.

At v_1 direction of greatest Attraction v_1

Repulsion v_2

Scanned with CamScanner

Day: M T W T F S Important short Examples Date: / /20

Rashid Hussain lect 31 to 35

(15) Suppose $B=\{b_1, b_2\}$ is a basis for V and $C=\{c_1, c_2, c_3\}$ is a basis for W. Let $T: V \to W$ be a Linear transformation with the property that

$$T(b_1) = 3c_1 - 2c_2 + 5c_3 \text{ and } T(b_2) = 4c_1 + 7c_2 - c_3$$

find the matrix M for T relative to B and C.

Ans Here $T(b_1)_C$ and $[T(b_2)]_C$

$$[T(b_1)]_C = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}, \quad T(b_2)_C = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$$

(16) find $T(a_0 + a_1 t + a_2 t^2)$, if T is the Linear transformation from P_2 to P_2 whose matrix relative to $B=\{1, t, t^2\}$ is $[T]_B = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix}$

Ans

$$[T(P)_B]_B = [T]_B [P]_B$$

$$= \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$= \begin{bmatrix} 3a_0 + 4a_1 \\ 5a_1 - a_2 \\ a_0 - 2a_1 + 7a_2 \end{bmatrix}$$

$$T(P) = (3a_0 + 4a_1)1 + (5a_1 - a_2)t + (a_0 - 2a_1 + 7a_2)t^2$$

Scanned with CamScanner

Day: M T W T F S Date: __/__/20

(12) let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. find a formula for A^k.
given that $A = PDP^{-1}$, where
$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

(13) determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

(sol) $\lambda_1, \lambda_2, \lambda_3$
eigen value are $\lambda = 5, 0, -2$ are diagonalizable by theorem 2

(14) Compute A^8 where $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$

(sol) first we find eigenvalues and eigenvector

$$P = \begin{bmatrix} \text{eigen} \\ \text{vector} \end{bmatrix}, \quad D = \begin{bmatrix} \text{eigen} \\ \text{value} \end{bmatrix} \quad \begin{matrix} \lambda_1 = 2 \\ \lambda_2 = 1 \end{matrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$A^8 = PD^8P^{-1} = ?$

Rashid Hussain

important example

lect 25 to 30

lect 23, 24 practice complete
All example

Scanned with CamScanner

MTh401

Day: M T W T F S Date: __/__/20

Q No. 4. Example (4) page # 363

Same Q.No. 3 previous

$X = c_1 X_1 + c_2 X_2$

(Solve yourself)

Q.No.5 Same

Exercise (4.2) page # 367

Q.No.3 $\frac{dx}{dt} = -3x + 4y - 9z$

$= 6x - y$

$= 10x + 4y + 3z$

let $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\frac{dx}{dt} = \begin{bmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & 4 & 3 \end{bmatrix} X$

$X' = \begin{bmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & 4 & 3 \end{bmatrix}$

QNo (4)

Scanned with CamScanner

(3)

$\sin\theta = -\cos\theta$

$\cos\theta = \sin\theta$

Day: M T W T F S

mth401

Date: / /20

Its given homogeneous system of diff. eq.

Q.No.3 Example (3) pag # 362 (Short question)

Solution: $\frac{dx}{dt} = c_1\frac{dx}{dt} = c_1\frac{d}{dt}\begin{bmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ \sin t - \cos t \end{bmatrix}$ — (1)

(Short Q)

$$AX = c_1\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{bmatrix}\begin{bmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{bmatrix}$$

$$= c_1\begin{bmatrix} \cos t + 0 - \cos t - \sin t \\ \cos t - \frac{1}{2}\cos t + \frac{1}{2}\sin t + 0 \\ -2\cos t + 0 + \cos t + \sin t \end{bmatrix}$$

$$= c_1\begin{bmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t + \sin t \end{bmatrix}$$

Rashid Hussain

$$AX = c_1\begin{bmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ \sin t - \cos t \end{bmatrix}$$ — (ii)

We have verified $\frac{dx}{dt} = AX$

Hence the vector c_1X_1 is also a solution vector of the homogeneous system of diff. eq.

Scanned with CamScanner

(2) Mth 401

Day: M T W T F S Date: / /20

Further, $AX_1 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$

$= \begin{pmatrix} e^{-2t} - 3e^{-2t} \\ 5e^{-2t} - 3e^{-2t} \end{pmatrix}$

$AX_1 = \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix} = X_1'$

(short) (بہت آسان ہے)

Similarly, $X_2 = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$

$X_2' = \begin{pmatrix} 18e^{6t} \\ 30e^{6t} \end{pmatrix}$

further, $AX_2 = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$

$= \begin{bmatrix} 3e^{6t} + 15e^{6t} \\ 15e^{6t} + 15e^{6t} \end{bmatrix}$

$= \begin{bmatrix} 18e^{6t} \\ 30e^{6t} \end{bmatrix} = AX_2'$

Rashid Hussain

The vectors X_1 and X_2 satisfy the homogeneous linear system

$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$

the given vectors are the solutions of the given

MTH401 - Important short questions for final term

Day: M T W T F S Date: / /20

Q.No.4. Find the general solution of the given diff eq

$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \left(x^2 - \frac{1}{64}\right)y = 0$ in $(0,\infty)$

Sol:- the Bessel's diff eq is

$x^2y'' \text{ or } x^2\frac{d^2y}{dx^2} + xy' + (x^2 - v^2)y = 0$ — (1)

Given equation is

$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \left(x^2 - \frac{1}{64}\right)y = 0$ — (2)

Comparing eq (1) and (2) we know that

$v^2 = \frac{1}{64}$ or $v^2 - \frac{1}{64} = 0$

$\left(v - \frac{1}{8}\right)\left(v + \frac{1}{8}\right) = 0$

$v - \frac{1}{8} = 0$ or $v + \frac{1}{8} = 0$

$v = \frac{1}{8}$ or $v = -\frac{1}{8}$

$v = \pm\frac{1}{8}$

General solution

$\therefore y = c_1J_{\frac{1}{8}}(x) + c_2J_{-\frac{1}{8}}(x)$

General solution is

$y = c_1J_{v=\frac{1}{8}}(x) + c_2J_{-\frac{1}{8}}(x)$

Q.No.5 Find the general solution of the given diff eq $\text{on } (0,\infty)$

$4x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + (4x^2 - 25) = 0$

Sol:- the Bessel's diff. equation is

$x^2y'' + xy'x + (x^2 - v^2)y = 0$ — (1)

Given diff. equation is

$4x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + (4x^2 - 25) = 0$

Diving by 4

Scanned with CamScanner

MTH401 (Very important)

Day: M T W T F S short Question Date: 1/120

lect 9
Q No 1 If a mas weighing 8 lb and k = 5 lb/ft
then find the amount of elongation's.

Sol:- weighing w = 8 lb S = ?
k = 5 lb/ft S = ? w = k l
we know that (Hook's Law · F = Ks $s = \frac{w}{k}$
w = ks w = kS
8 lb = 5 lb/ft · S
$\frac{8}{5}$ slugs $\frac{ft}{ft}$ = S
$\frac{8}{5}$ ft = S

lect #40
Q No 2. write the following system in matrix form

$$\frac{dx}{dt} = -3x + 4y + e^{-t}\sin 2t$$

$$\frac{dy}{dt} = 5x + 9y + 4e^{-t}\cos 2t$$

Sol:

$$\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} -3 & 4 \\ 5 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} \sin 2t \\ 4\cos 2t \end{bmatrix} e^{-t}$$

lect #40
Q No 3 write the system $\frac{dx}{dt} = 6x + y + 6t$,
$\frac{dy}{dt} = 4x + 3y - 10t + 4$ in the form of $x' = Ax + f(t)$

Sol:-

$$\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 6t \\ -10t + 4 \end{bmatrix}$$

$$x' = Ax + f(t)$$

Scanned with CamScanner

Day: M T W T F S Date: / /20

$$\frac{4x^2}{4}\frac{d^2y}{dx^2} + \frac{4x}{x}\frac{dy}{dx} + \left(\frac{4x^2}{4} - \frac{25}{4}\right)y = 0$$

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \left(x^2 - \frac{25}{4}\right)y = 0 \quad -(2)$$

Comparing Equation (1) and (2), we get

$$v^2 = \frac{25}{4}$$

$$v^2 - \frac{25}{4} = 0$$ Rashid Hussain

$$\left(v - \frac{5}{4}\right)\left(v + \frac{5}{4}\right) = 0$$

$$v - \frac{5}{4} = 0 \quad \text{or} \quad v + \frac{5}{4} = 0$$

$$v = 5/4 \quad \text{or} \quad v = -5/4$$

$$2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$$

$$y = c_1 J_v(x) + c_2 J_{-v}(x)$$

So the general solution is

$$y = c_1 J_v(x) + c J_{-v}(x)$$

$$= c_1 J_{5/4}(x) + c J_{-5/4}(x)$$

Q.No. (6) Solve the diff. equation

$$\frac{dy}{dx} = \frac{x^2}{2y}$$ (Not in syllabus but solve it better)

Sol:-

$$2y\,dy = x^2\,dx$$

$$\int 2y\,dy = \int x^2\,dx$$

$$\frac{2y^2}{2} = \frac{x^3}{3} + c$$

$$y^2 = \frac{x^3}{3} + c$$ Ans

Scanned with CamScanner

Day: M T W T F S MTH 401 Short question for final term Date: __/__/20__

Q.No.7 Find the complementary solution of D.E

$$y'' + 4y' + 4y = 2e^{2x}$$

Sol:- $D^2y + 4Dy + 4D = 2e^{2x}$

To find the complementary solution, y_c, we consider the associate homogeneous diff. eq.

$$(D^2 + 4D + 4)y = 0$$

$$m^2 + 4m + 4 = 0$$

$$(m+2)^2 = 0$$

$m_1 = -2, m_2 = -2$ $\quad y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

$$\therefore y_c = c_1 e^{-2x} + c_2 x e^{-2x}$$

$$= (c_1 + c_2 x) e^{-2x}$$ Ans

$$y_c = (c_1 + c_2 x) e^{-2x}$$

Q.No.8

Scanned with CamScanner

Date: / /20

(5) Power Series Solution of a Differential Eq.

$\frac{dy}{dx} - 2xy = 0$

Solution in power series

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

(6) Ordinary and singular point

A point x_0 is said to be a ordinary point if diff. eq $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ if both $P(x)$ and $Q(x)$ are analytical at x_0.

Rashid Hussain

singular point

A point that is not an ordinary point is said to be singular point of the equation.

(7) Solution about singular point

if $x = x_0$ is singular point, it is not always possible to find a solution of the form

$$y = \sum_{n=0}^{\infty} C_n (x - x_0)^n \text{ for the equation}$$

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

(8) Regular and irregular Singular points

Regular singular points: A singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a regular singular point if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0.

Scanned with CamScanner

(1) Theortical Short MTH401

Day: M T W T F S Date: / /20

(1) Cauchy - Euler Equation. Rashid Hussain

Any Linear differential Eq. of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots\cdots a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

where $a_n, a_{n-1}, \ldots\ldots a_1, a_0$ are Constant is said to be a Cauchy-Euler equation or equi dimensional eq.

(2) Power Series.

A Power series in $(x-a)$ is an infinite series of the form

$$\sum_{n=0}^{\infty} C_n (x-a)^n = C_0 + C_1 (x-a) + C_2 (x-a)^2 + \cdots\cdots$$

$C_0, C_1, C_2, \ldots$ are Constant on x represent a variable.

(3) Analytic at a Point

A function f is said to be analytic at point a if the function can be represented by power series in $(x-a)$ with a positive radius of convergence.

(4) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\cdots$

Rashid Hussain

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots\cdots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots\cdots$$

$$\sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \frac{61x^6}{720} + \cdots\cdots$$

$$e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \cdots\cdots$$

Scanned with CamScanner

(13)

Irregular Singular Points

Rashid Hossain

A singular point that is not regular is said to be an irregular singular point of the equation.

(9) frobenius' theorem

if $x = x_0$ is a regular singular point of the eq. $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one series solution of the form

$$y = (x-x_0)^r \sum_{n=0}^{\infty} c_n (x-x_0)^n = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r}$$

(10) Bessel's Diff. Eq.

A second order Linear diff. eq. of the form

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - v^2)y = 0$$

is called Bessel's diff. eq. $J_v(x)$ Bessel function

Bessel's function

(11) Bessel's function of first Kind

$r_1 = v$

$$J_v(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v}$$

$r_2 = -v$

$$J_{-v}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1-v+n)} \left(\frac{x}{2}\right)^{2n-v}$$

The function $J_v(x)$ and $J_{-v}(x)$ are called Bessel functions.

(Next Commin Soon)

Scanned with CamScanner

Day: M T W T F Date: / /20

MTH401 Rashid

(12) Spherical Bessel function

Bessel function of index half and an odd integer are called spherical Bessel function. Like other Bessel function. Spherical Bessel functions are used in many physical problems.. Quiz

(13) Lengendre's Differential Eq.

A second order linear differentialeq of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's diff. eq. and any solution is called **Legendre's function.**

(14) Rodrigue formula for Legendre's polynomil

The legendre polynomials are also generated by Rodrigue formula

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$$

Quiz Rashid Hossain

(15) Degenerate System

the system of differential equation of the

$$P_{n1}(D)x_1 + P_{n2}(D)x_2 + \cdots P_{nn}(D)x_n = b_n(t)$$

those cannot be reduced to a linear system in normal form is said to be a **degenerate system.**

→ to the canonical form.

system Reduced to normal form

normal

Day: S M T W T F S Date: / /20

(16) Derivative of a matrix of function

Suppose that

$$A(t) = [a_{ij}(t)]_{m\times n}$$

Derivative of the matrix $A(t)$ is a matrix whose entries are derivatives of the corresponding entries of the matrix $A(t)$. Then

$$\frac{dA}{dt} = \left[\frac{d}{dt}a_{ij}\right]_{m\times n}$$

Rashid Hussain

The derivative of a matrix is also denoted by $\dot{A}(t)$

(17) Integral of a matrix of functions

Let $A(t) = (a_{ij}(t))_{m\times n}$, $\int_{t_0}^{t} A(s)\,ds = \left[\int_{t_0}^{t} a_{ij}(s)\,ds\right]_{m\times n}$

Rashid Hussain

(18) Notation

symbol	meaning
R_{ij}	interchange the rows i and j
cR_i	multiply the ith row by a nonzero constant c.
$cR_i + R_j$	multiply the ith row by c and then add to the jth row.

Scanned with CamScanner

MTH401

Day: M T W T F S Date: __/__/20

Red lect 23,24

imp. points

(1) Damping force $= -B\left(\frac{dx}{dt}\right)$

(2) Restoring force $= -K(S+x)$

(3) The coefficient $Ae^{-\lambda t}$ is called damped amplitude of vibrations.

(4) $\frac{2\pi}{\sqrt{\omega^2-\lambda^2}}$ quasi period

(5) $\frac{\sqrt{\omega^2-\lambda^2}}{2\pi}$ quasi frequency

Rashid Hussain

(6) **force motion**

A vibration system consisting of a body of mass m attached to a spring. the motion of the body is being driven by an external force $f(t)$.

Scanned with CamScanner

Day: S M T W T F S Date: / / 20

(19) Eigenvalue and Eigenvectors

Let A be a nxn matrix. A number λ is said to be an eigenvalue of A if there exists a nonzero solution vector K of the system of Linear diff eq. $AK = \lambda K$

the solution vector K is said to be an eigen vector.

(20) The non-trivial solution.

The non-trival solution of the system exists only when

$$\det(A - \lambda I) = 0$$

(21) Initial-Value problem.

Let to denote any point in some interval denote by I and

$$X(t_0) = \begin{bmatrix} x_1(t_0) \\ x_2(t_0) \\ \vdots \\ x_n(t_0) \end{bmatrix}, \quad X_0 = \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_n \end{bmatrix}$$

Rashid Hussain

Diff. eq vector

$$\frac{dx}{dt} = A(t)X + f(t)$$

initial condition

$$X(t_0) = X_0$$

is called an initial problem on the interval I.

Scanned with CamScanner

29 Power Series Solution of a Differential Equation

We know that the explicit solution of the linear first-order differential equation

$$\frac{dy}{dx} - 2xy = 0$$

$$y = e^{x^2}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

Also

If we replace x by x^2 in the series representation of e^x, we can write the solution of the differential equation as

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

This last series converges for all real values of x. In other words, knowing the solution in advance, we were able to find an infinite series solution of the differential equation.

We now propose to obtain a **power series solution** of the differential equation directly; the method of attack is similar to the technique of undetermined coefficients.

Example 11

Find a solution of the DE: $\frac{dy}{dx} - 2xy = 0$ in the form of power series in x.

Solution If we assume that a solution of the given equation exists in the form

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + \sum_{n=1}^{\infty} c_n x^n$$

The question is that: Can we determine coefficients c_n for which the power series converges to a function satisfying the differential equation? Now term-by-term differentiation of the proposed series solution gives

$$\frac{dy}{dx} = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

Using the last result and the assumed solution, we have

$$\frac{dy}{dx} - 2xy = \sum_{n=1}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} 2 c_n x^{n+1}$$

We would like to add the two series in this equation. To this end we write

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$\frac{dy}{dx} - 2xy = 1 \cdot c_1 x^0 + \sum_{n=2}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$ and then proceed as in the previ

example by letting $k = n-1, \quad k = n+1$ in the first and second series, respective

Therefore, last equation becomes $\frac{dy}{dx} - 2xy = c_1 + \sum_{k=1}^{\infty} (k+1) c_{k+1} x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k$

After we add the series term wise, it follows that

$$\frac{dy}{dx} - 2xy = c_1 + \sum_{k=1}^{\infty} \left[(k+1) c_{k+1} - 2c_{k-1} \right] x^k$$

Substituting in the given differential equation, we obtain

$$c_1 + \sum_{k=1}^{\infty} \left[(k+1) c_{k+1} - 2c_{k-1} \right] x^k = 0$$

In order to have this true, it is necessary that all the coefficients must be zero. This me
that

$$c_1 = 0, \quad (k+1) c_{k+1} - 2c_{k-1} = 0, \qquad k = 1, 2, 3, \cdots$$

This equation provides a recurrence relation that determines the coefficient c_k. Si
$k + 1 \neq 0$ for all the indicated values of k, we can write as

$$c_{k+1} = \frac{2c_{k-1}}{k+1}$$

Iteration of this last formula then gives

$$k = 1, \qquad c_2 = \frac{2}{2} c_0 = c_0$$

$$k = 2, \qquad c_3 = \frac{2}{3} c_1 = 0$$

$$k = 3, \qquad c_4 = \frac{2}{4} c_2 = \frac{1}{2} c_0 = \frac{1}{2!} c_0$$

$$k = 4, \qquad c_5 = \frac{2}{5} c_3 = 0$$

$$k = 5, \qquad c_6 = \frac{2}{6} c_4 = \frac{1}{3 \cdot 2!} c_0 = \frac{1}{3!} c_0$$

$$k = 6, \qquad c_7 = \frac{2}{7} c_5 = 0$$

Scanned with CamScanner

30 Solution about Ordinary Points

30.1 Analytic Function

Quizes or short

A function f is said to be analytic at a point a if it can be represented by a power series in $(x-a)$ with a positive radius of convergence.

Suppose the linear second-order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \qquad (1)$$

is put into the form

$$y'' + P(x)y' + Q(x)y = 0 \qquad (2)$$

by dividing by the leading coefficient $a_2(x)$.

30.2 Ordinary and singular points

Shor and Quiz

A point x_0 is said to be a *ordinary point* of a differential equation (1) if both $P(x)$ and $Q(x)$ are analytic at x_0. A point that is not an ordinary point is said to be ***singular point*** of the equation.

30.2.1 Polynomial Coefficients

If $a_2(x), a_1(x)$ and $a_0(x)$ are polynomials with no common factors, then $x = x_0$ is

(i) an ordinary point if $a_2(x) \neq 0$ or

(ii) a singular point if $a_2(x) = 0$.

Example

Quiz and short

(a) The singular points of the equation $(x^2-1)y'' + 2xy' + 6y = 0$ are the solutions of $x^2 - 1 = 0$ or $x = \pm 1$. All other finite values of x are the ordinary points.

(b) The singular points need not be real numbers.

The equation $(x^2+1)y'' + 2xy' + 6y = 0$ has the singular points at the solutions of $x^2 + 1 = 0$, namely $x = \pm i$.

All other finite values, real or complex, are ordinary points.

Example The Cauchy-Euler equation $ax^2y'' + bxy' + cy = 0$, where a, b and c are constants, has singular point at $x = 0$.

All other finite values of x, real or complex, are ordinary points.

Quiz

Scanned with CamScanner

short and Quiz

30.3 Theorem (Existence of Power Series Solution)

If $x = x_0$ is an ordinary point of the differential equation $y'' + P(x)y' + Q(x)y = 0$, we can always find two linearly independent solutions in the form of power series centered at x_0: $y = \sum_{n=0}^{\infty} c_n (x - x_0)^n$. A series solution converges at least for $|x - x_0| < R$, where R is the distance from x_0 to the closest singular point (real or complex).

Example Solve $y'' - 2xy = 0$.

Solution

We see that $x = 0$ is an ordinary point of the equation. Since there are no finite singular points, there exist two solutions of the form $y = \sum_{n=0}^{\infty} c_n x^n$ convergent for $|x| < \infty$.

Proceeding, we write

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$y'' - 2xy = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

$$= 2 \cdot 1 c_2 x^0 + \sum_{n=3}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} 2c_n x^{n+1}$$

both series start with x^1

Letting $k = n - 2$ in the first series and $k = n + 1$ in the second, we have

$$y'' - 2xy = 2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k$$

$$= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1) c_{k+2} - 2c_{k-1}] x^k = 0$$

$$2c_2 = 0 \quad \text{and} \quad (k+2)(k+1) c_{k+2} - 2c_{k-1} = 0$$

The last expression is same as

$$c_{k+2} = \frac{2c_{k-1}}{(k+2)(k+1)}, \quad k = 1, 2, 3, \cdots$$

Scanned with CamScanner

Iteration gives $c_3 = \frac{2c_0}{3\cdot 2}$

$$c_4 = \frac{2c_1}{4\cdot 3}$$

$$c_5 = \frac{2c_2}{5\cdot 4} = 0 \qquad \text{because } c_2 = 0$$

$$c_6 = \frac{2c_3}{6\cdot 5} = \frac{2^2}{6\cdot 5\cdot 3\cdot 2}c_0$$

$$c_7 = \frac{2c_4}{7\cdot 6} = \frac{2^2}{7\cdot 6\cdot 4\cdot 3}c_1$$

$$c_8 = \frac{2c_5}{8\cdot 7} = 0$$

$$c_9 = \frac{2c_6}{9\cdot 8} = \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}c_0$$

$$c_{10} = \frac{2c_7}{10\cdot 9} = \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}c_1$$

$$c_{11} = \frac{2c_8}{11\cdot 10} = 0 \qquad \text{, and so on.}$$

It is obvious that both c_0 and c_1 are arbitrary. Now

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + c_{11}x^{11} + \cdots$$

$$y = c_0 + c_1x + 0 + \frac{2}{3\cdot 2}c_0x^3 + \frac{2}{4\cdot 3}c_1x^4 + 0 + \frac{2^2}{6\cdot 5\cdot 3\cdot 2}c_0x^6 + \frac{2^2}{7\cdot 6\cdot 4\cdot 3}c_1x^7 + 0$$
$$+ \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}c_0x^9 + \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}c_1x^{10} + 0 + \cdots$$

$$y = c_0\left[1 + \frac{2}{3\cdot 2}x^3 + \frac{2^2}{6\cdot 5\cdot 3\cdot 2}x^6 + \frac{2^3}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}x^9 + \cdots\right]$$
$$+ c_1\left[x + \frac{2}{4\cdot 3}x^4 + \frac{2^2}{7\cdot 6\cdot 4\cdot 3}x^7 + \frac{2^3}{10\cdot 9\cdot 7\cdot 6\cdot 4\cdot 3}x^{10} + \cdots\right].$$

Scanned with CamScanner

$$c_3 = -\frac{2}{5}c_3 = 0$$

$$c_6 = -\frac{3}{6}c_4 = \frac{3}{2\cdot 4\cdot 6}c_0 = \frac{1\cdot 3}{2^3 3!}c_0$$

$$c_7 = -\frac{4}{7}c_5 = 0$$

$$c_8 = -\frac{5}{8}c_6 = -\frac{3\cdot 5}{2\cdot 4\cdot 6\cdot 8}c_0 = -\frac{1\cdot 3\cdot 5}{2^4 4!}c_0$$

$$c_9 = -\frac{6}{9}c_7 = 0$$

$$c_{10} = -\frac{7}{10}c_8 = -\frac{3\cdot 5\cdot 7}{2\cdot 4\cdot 6\cdot 8\cdot 10}c_0 = \frac{1\cdot 3\cdot 5\cdot 7}{2^5 5!}c_0 \text{ and so on.}$$

Therefore

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + c_7 x^7 + c_8 x^8 + \cdots$$

$$y = c_1 x + c_0\left[1 + \frac{1}{2}x^2 - \frac{1}{2^2 2!}x^4 + \frac{1\cdot 3}{2^3 3!}x^6 - \frac{1\cdot 3\cdot 5}{2^4 4!}x^8 + \frac{1\cdot 3\cdot 5\cdot 7}{2^5 5!}x^{10} - \cdots\right]$$

The solutions are

$$y_1(x) = c_0\left[1 + \frac{1}{2}x^2 + \sum_{n=2}^{\infty}(-1)^{n-1}\frac{1\cdot 3\cdot 5\cdots(2n-3)}{2^n n!}x^{2n}\right], \quad |x| < 1$$

$$y_2(x) = c_1 x.$$

Example If we seek a solution $y = \sum_{n=0}^{\infty} c_n x^n$ for the equation $y'' - (1+x)y = 0$,

we obtain $c_2 = c_0/2$ and the three-term recurrence relation

$$c_{k+2} = \frac{c_k + c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \cdots$$

To simplify the iteration we can first choose $c_0 \neq 0, c_1 = 0$; this yields one solution. The other solution follows from next choosing $c_0 = 0, c_1 \neq 0$. With the first assumption we find

© Copyright Virtual University

Scanned with CamScanner

Example Solve $(x^2+1)y'' + xy' - y = 0.$

Solution

Since the singular points are $x = \pm i$, $x = 0$ is the ordinary point, a power series will converge at least for $|x| < 1$. The assumption $y = \sum_{n=0}^{\infty} c_n x^n$ leads to

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

$$= \sum_{n=2}^{\infty} n(n-1)c_n x^n + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} nc_n x^n - \sum_{n=0}^{\infty} c_n x^n$$

$$= 2c_2 x^0 - c_0 x^0 + 6c_3 x + c_1 x - c_1 x + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + \underbrace{\sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=2}^{\infty} nc_n x^n}_{k=n} - \underbrace{\sum_{n=2}^{\infty} c_n x^n}_{k=n}$$

$$= 2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [k(k-1)c_k + (k+2)(k+1)c_{k+2} + kc_k - c_k]x^k = 0$$

or $$2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2}]x^k = 0.$$

Thus $$2c_2 - c_0 = 0$$

$$c_3 = 0$$

$$(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2} = 0$$

This implies

$$c_2 = \frac{1}{2}c_0$$

$$c_3 = 0$$

$$c_{k+2} = \frac{-(k-1)}{(k+2)}c_k, \quad k = 2, 3, \cdots$$

Iteration of the last formula gives

$$c_4 = -\frac{1}{4}c_2 = -\frac{1}{2\cdot 4}c_0 = -\frac{1}{2^2 2!}c_0$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$c_1 = \frac{1}{2}c_0$

$$c_3 = \frac{c_1 + c_2}{2 \cdot 3} = \frac{c_0}{2 \cdot 3} = \frac{1}{6}c_0$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{c_0}{2 \cdot 3 \cdot 4} = \frac{1}{24}c_0$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{c_0}{4 \cdot 5}\left[\frac{1}{2 \cdot 3} + \frac{1}{2}\right] = \frac{1}{30}c_0 \text{ and so on.}$$

Thus one solution is

$$y_1(x) = c_0\left[1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{30}x^5 + \cdots\right].$$

Similarly if we choose $c_0 = 0$, then

$$c_2 = 0$$

$$c_3 = \frac{c_1 + c_2}{2 \cdot 3} = \frac{c_1}{2 \cdot 3} = \frac{1}{6}c_1$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{c_1}{3 \cdot 4} = \frac{1}{12}c_1$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{c_1}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{120}c_1 \text{ and so on.}$$

Hence another solution is $y_2(x) = c_1\left[x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{120}x^5 + \cdots\right]$.

Each series converges for all finite values of x.

30.4 Non-polynomial Coefficients

The next example illustrates how to find a power series solution about an ordinary point of a differential equation when its coefficients are not polynomials. In this example we see an application of multiplication of two power series that we discussed earlier.

Example Solve $y'' + (\cos x)y = 0$

Solution The equation has no singular point. half short and Quiz

Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$, it is seen that $x = 0$ is an ordinary point. Quiz

Thus the assumption $y = \sum_{n=0}^{\infty} c_n x^n$ leads to

Scanned with CamScanner

$$y'' + (\cos x)y = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots)\sum_{n=0}^{\infty} c_n x^n$$

$$= (2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \cdots) + (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots)(c_0 + c_1 x + c_2 x^2 \ldots$$

$$= 2c_2 + c_0 + (6c_3 + c_1)x + (12c_4 + c_2 - \frac{1}{2}c_0)x^2 + (20c_5 + c_3 - \frac{1}{2}c_1)x^3 + \cdots$$

If the last line be identically zero, we must have

$$2c_2 + c_0 = 0 \Rightarrow c_2 = -\frac{c_0}{2}$$

$$6c_3 + c_1 = 0 \Rightarrow c_3 = -\frac{c_1}{6}$$

$$12c_4 + c_2 - \frac{1}{2}c_0 = 0 \Rightarrow c_4 = \frac{c_0}{12}$$

$$20c_5 + c_3 - \frac{1}{2}c_1 = 0 \Rightarrow c_5 = \frac{c_1}{30}$$ and so on. c_0 and c_1 are arbitrary.

Now $y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \cdots$

or $$y = c_0 + c_1 x - \frac{c_0}{2}x^2 - \frac{c_1}{6}x^3 + \frac{c_0}{12}x^4 + \frac{c_1}{30}x^5 - \cdots$$

$$y = c_0(1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \cdots) + c_1(x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \cdots)$$

$y_1(x) = c_0[1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \cdots]$ and $y_2(x) = c_1[x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \cdots]$

Since the differential equation has no singular point, both series converge for all finite values of x.

30.5 Exercise

In each of the following problems find two linearly independent power series solutions about the ordinary point $x = 0$.

1. $y'' + x^2 y = 0$
2. $y'' - xy' + 2y = 0$
3. $y'' + 2xy' + 2y = 0$
4. $(x+2)y'' + xy' - y = 0$
5. $(x^2 + 2)y'' - 6y = 0$

Scanned with CamScanner

31 Solutions about Singular Points

short

If $x = x_0$ is singular point, it is not always possible to find a solution of the form $y = \sum_{n=0}^{\infty} c_n (x - x_0)^n$ for the equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$

Quiz

However, we may be able to find a solution of the form

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$, where r is constant to be determined.

To define regular/irregular singular points, we put the given equation into the standard form $y'' + P(x)y' + Q(x)y = 0$

31.1 Regular and Irregular Singular Points

short Quiz

A Singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a ***regular singular point*** if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0. A singular point that is not regular is said to be an ***irregular singular point*** of the equation.

31.1.1 Polynomial Coefficients

If the coefficients in the given differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ are polynomials with no common factors, above definition is equivalent to the following:

*Let $a_2(x_0) = 0$ Form $P(x)$ and $Q(x)$ by reducing $\frac{a_1(x)}{a_2(x)}$ and $\frac{a_0(x)}{a_2(x)}$ to lowest terms, respectively. If the factor $(x - x_0)$ appears at most to the first powers in the denominator of $P(x)$ and at most to the second power in the denominator of $Q(x)$, then $x = x_0$ is a **regular singular point**.*

Example 1 $x = \pm 2$ are singular points of the equation $(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$

short and Quiz

Dividing the equation by $(x^2 - 4)^2 = (x - 2)^2 (x + 2)^2$, we find that

$$P(x) = \frac{1}{(x - 2)(x + 2)^2} \quad \text{and} \quad Q(x) = \frac{1}{(x - 2)^2 (x + 2)^2}$$

1. $x = 2$ is a regular singular point because power of $x - 2$ in $P(x)$ is 1 and in $Q(x)$ is 2. (is not = ∞)
2. $x = -2$ is an irregular singular point because power of $x + 2$ in $P(x)$ is 2. (= ∞)

The 1st condition is violated.

Example 2 short and long

Both $x = 0$ and $x = -1$ are singular points of the differential equation.

Scanned with CamScanner

$$x\,y'' - 2xy' + 5y = 0$$

Because the equation can be written as $y'' - 2y' + \frac{5}{x}y = 0$ giving $P(x) = -2$ and $Q(x) = \frac{5}{x}$.

In part (c) of Example 3 we noticed that $(x-0)$ and $(x-0)^2$ do not even appear in the denominators of $P(x)$ and $Q(x)$ respectively. Remember, these factors can appear at most in this fashion. For a singular point $x = x_0$, any nonnegative power of $(x - x_0)$ less than one (namely, zero) and nonnegative power less than two (namely, zero and one) in the denominators of $P(x)$ and $Q(x)$, respectively, imply x_0 is a ***regular singular point.***

Please note that the singular points can also be complex numbers.

For example, $x = \pm 3i$ are regular singular points of the equation

$$(x^2+9)y'' + -3xy' + (1-x)y = 0$$

Because the equation can be written as

$$y'' - \frac{3x}{x^2+9}y' + \frac{1-x}{x^2+9}y = 0.$$

$$\therefore P(x) = \frac{-3x}{(x-3i)(x+3i)}, \quad Q(x) = \frac{1-x}{(x-3i)(x+3i)}.$$

31.2 Method of Frobenius

To solve a differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ about a regular singular point we employ the Frobenius' Theorem.

31.2.1 Frobenius' Theorem

If $x = x_0$ is a regular singular point of equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one series solution of the form

$$y = (x-x_0)^r \sum_{n=0}^{\infty} c_n (x-x_0)^n = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r}$$

where the number r is a constant that must be determined. The series will converge at least on some interval $0 < x - x_0 < R$.

Note that the solutions of the form $y = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r}$ are not guaranteed.

Method of Frobenius

1. Identify regular singular point x_0.

© Copyright Virtual University of Pak

Scanned with CamScanner

and $$c_{k+1} = \frac{c_k}{(k+r+1)(3k+3r+1)}, k = 0,1,2,\ldots$$

Substitute $r_1 = \frac{2}{3}$ and $r_2 = 0$ in the above equation and these values will give two different recurrence relations:

For $r_1 = \frac{2}{3}$, $$c_{k+1} = \frac{c_k}{(3k+5)(k+1)}, \quad k = 0,1,2,\ldots \quad (1)$$

For $r_2 = 0$ $$c_{k+1} = \frac{c_k}{(k+1)(3k+1)}, \quad k = 0,1,2,\ldots \quad (2)$$

Iteration of (1) gives

$$c_1 = \frac{c_0}{5.1}$$

$$c_2 = \frac{c_1}{8.2} = \frac{c_0}{2!5.8}$$

$$c_3 = \frac{c_2}{11.3} = \frac{c_0}{3!5.8.11}$$

$$c_4 = \frac{c_3}{14.4} = \frac{c_0}{4!5.8.11.14}$$

In general $$c_n = \frac{c_0}{n!5.8.11.14\ldots(3n+2)}, \quad n = 1,2,\ldots$$

Iteration of (2) gives

$$c_1 = \frac{c_0}{1.1}$$

$$c_2 = \frac{c_1}{2.4} = \frac{c_0}{2!1.4}$$

$$c_3 = \frac{c_2}{3.7} = \frac{c_0}{3!1.4.7}$$

$$c_4 = \frac{c_3}{4.10} = \frac{c_0}{4!1.4.7.10}$$

In general $$c_n = \frac{c_0}{n!1.4.7\ldots(3n-2)}, \quad n = 1,2,\ldots$$

Thus we obtain two series solutions

Scanned with CamScanner

2. Substitute $y=\sum_{n=0}^{\infty} c_n(x-x_0)^{n+r}$ in the given differential equation,
3. Determine the unknown exponent r and the coefficients c_n.
4. For simplicity assume that $x_0=0$.

Example 4

As $x=0$ is regular singular points of the differential equation

$$3xy''+y'-y=0.$$

We try a solution of the form $y=\sum_{n=0}^{\infty} c_n x^{n+r}$.

Therefore $y'=\sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1}$.

And $y''=\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-2}$.

$$3xy''+y'-y=3\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1}+\sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1}-\sum_{n=0}^{\infty}c_n x^{n+r}$$

$$=\sum_{n=0}^{\infty}(n+r)(3n+3r-2)c_n x^{n+r-1}-\sum_{n=0}^{\infty}c_n x^{n+r}$$

$$=x^r\left[r(3r-2)c_0x^{-1}+\sum_{n=1}^{\infty}(n+r)(3n+3r-2)c_n x^{n-1}-\sum_{n=0}^{\infty}c_n x^n\right]$$

$$=x^r\left[r(3r-2)c_0x^{-1}+\sum_{k=0}^{\infty}[(k+r+1)(3k+3r+1)c_{k+1}-c_k]x^k\right]=0$$

$k=n-1$ $\quad k=n$

which implies $r(3r-2)c_0=0$

$(k+r+1)(3k+3r+1)c_{k+1}-c_k=0$, $k=0,1,2,\ldots$

Since nothing is gained by taking $c_0=0$, we must then have

$r(3r-2)=0$ [called the indicial equation and its roots $r=\frac{2}{3},0$ are called indicial roots or exponents of the singularity.]

© Copyright Virtual Univ

Scanned with CamScanner

$$c_1 = \frac{c_0}{1.3}$$

$$c_2 = \frac{c_1}{2.4} = \frac{2c_0}{2!4!}$$

$$c_3 = \frac{c_2}{3.5} = \frac{2c_0}{3!5!}$$

$$c_4 = \frac{c_3}{4.6} = \frac{2c_0}{4!6!}$$

$$\vdots$$

$$c_n = \frac{2c_0}{n!(n+2)!}, \quad n = 1, 2, \ldots$$

Thus one series solution is

$$y_1 = c_0 x^0 \left[1 + \sum_{n=1}^{\infty} \frac{2}{n!(n+2)!} x^n \right] = c_0 \sum_{n=0}^{\infty} \frac{2}{n!(n+2)!} x^n, |x| < \infty.$$

Now when $r_2 = -2$, (1) becomes

$$(k-1)(k+1)c_{k+1} - c_k = 0 \qquad (2)$$

but note here that we do not divide by $(k-1)(k+1)$ immediately since this term is zero for $k = 1$. However, we use the recurrence relation (2) for the cases $k = 0$ and $k = 1$:

$$-1.1c_1 - c_0 = 0 \qquad \text{and} \quad 0.2c_2 - c_1 = 0$$

The latter equation implies that $c_1 = 0$ and so the former equation implies that $c_0 = 0$.

Continuing, we find

$$c_{k+1} = \frac{c_k}{(k-1)(k+1)} \qquad k = 2, 3, \ldots$$

$$c_3 = \frac{c_2}{1.3}$$

$$c_4 = \frac{c_3}{2.4} = \frac{2c_2}{2!4!}$$

$$c_5 = \frac{c_4}{3.5} = \frac{2c_2}{3!5!}, \ldots$$

$$\vdots$$

Scanned with CamScanner

38 Introduction to Matrices

38.1 Matrix

A rectangular array of numbers or functions subject to certain rules and conditions is called a matrix. Matrices are denoted by capital letters $A, B, \ldots, Y, Z$. The numbers or functions are called elements or entries of the matrix. The elements of a matrix are denoted by small letters $a, b, \ldots, y, z$.

38.2 Rows and Columns

The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

38.3 Order of a Matrix

If a matrix has m rows and n columns then we say that the size or order of the matrix is $m \times n$. If A is a matrix having m rows and n columns then the matrix can be written as

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

38.4 Square Matrix

A matrix having n rows and n columns is said to be a $n \times n$ square matrix or a square matrix of order n. The element, or entry, in the ith row and jth column of a $m \times n$ matrix A is written as a_{ij}. Therefore a 1 x 1 matrix is simply a constant or a function.

38.5 Equality of matrix

Any two matrices A and B are said to be equal if and only if they have the same orders and the corresponding elements of the two matrices are equal. Thus if $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then

$$A = B \Leftrightarrow a_{ij} = b_{ij}, \quad \forall\, i, j$$

38.6 Column Matrix

A column matrix X is any matrix having n rows and only one column. Thus the column matrix X can be written as

Scanned with CamScanner

In general $c_n = \dfrac{2c_2}{(n-2)!n!}, \quad n = 3, 4, 5, \ldots$

Thus $y_2 = c_2 x^{-2}\left[x^2 + \sum_{n=3}^{\infty} \dfrac{2}{(n-2)!n!} x^n\right]. \qquad (3)$

However, close inspection of (3) reveals that y_2 is simply constant multiple of

To see this, let $k = n-2$ in (3). We conclude that the method of Frobenius gives the series solution of the given differential equation.

31.3 Cases of Indicial Roots

When using the method of Frobenius, we usually distinguish three cases corresponding the nature of the indicial roots. For the sake of discussion let us suppose that r_1 and r_2 the real solutions of the indicial equation and that, when appropriate, r_1 denotes largest root.

31.3.1 Case I (Roots not Differing by an Integer)

If r_1 and r_2 are distinct and do not differ by an integer, then their exist two linearly independent solutions of the differential equation of the form

$$y_1 = \sum_{n=0}^{\infty} c_n x^{n+r_1}, \; c_0 \neq 0, \text{ and } y_2 = \sum_{n=0}^{\infty} b_n x^{n+r_2}, \quad b_0 \neq 0.$$

Example 6 Solve $2xy'' + (1+x)y' + y = 0.$

Solution: If $y = \sum_{n=0}^{\infty} c_n x^{n+r}$, then

$$2xy'' + (1+x)y' + y = 2\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$= \sum_{n=0}^{\infty}(n+r)(2n+2r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r+1)c_n x^{n+r}$$

$$= x^r\left[r(2r-1)c_0 x^{-1} + \sum_{n=1}^{\infty}(n+r)(2n+2r-1)c_n x^{n-1} + \sum_{n=0}^{\infty}(n+r+1)c_n x^n\right]$$

$$= x^r\left[r(2r-1)c_0 x^{-1} + \sum_{k=0}^{\infty}[(k+r+1)(2k+2r+1)c_{k+1} + (k+r+1)c_k]x^k\right] = 0$$

($n = k+1$; $k = n$)

which implies $r(2r-1) = 0$

$$(k+r+1)(2k+2r+1)c_{k+1} + (k+r+1)c_k = 0, \; k = 0, 1, 2, \ldots \qquad (1)$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

38.10 Multiplication of Matrices

Any two matrices A and B are conformable for the product AB, if the number of columns in the first matrix A is equal to the number of rows in the second matrix B. Thus if the order of the matrix A is $m \times n$ then to make the product AB possible order of the matrix B must be $n \times p$. Then the order of the product matrix AB is $m \times p$. Thus

$$A_{m\times n} \cdot B_{n\times p} = C_{m\times p}$$

If the matrices A and B are given by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

$$= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \cdots + a_{1n}b_{n1} \cdots a_{11}b_{1p} + a_{12}b_{2p} + \cdots + a_{1n}b_{np} \\ a_{21}b_{11} + a_{22}b_{21} + \cdots + a_{2n}b_{n1} \cdots a_{21}b_{1p} + a_{22}b_{2p} + \cdots + a_{2n}b_{np} \\ \vdots \\ \vdots \\ a_{m1}b_{11} + a_{m2}b_{21} + \cdots + a_{mn}b_{n1} \cdots a_{m1}b_{1p} + a_{m2}b_{2p} + \cdots + a_{mn}b_{np} \end{pmatrix}$$

$$= \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{n\times p}$$

Example 4

If possible, find the products AB and BA, when

(a) $$A = \begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}$$

Scanned with CamScanner

38.8 Addition of Matrices

Any two matrices can be added only when they have same orders and the resulting matrix is obtained by adding the corresponding entries. Therefore, if $A=[a_{ij}]$ and $B=[b_{ij}]$ are two $m\times n$ matrices then their sum is defined to be the matrix $A+B$ defined by $A+B=[a_{ij}+b_{ij}]$

Example 2 Consider the following two matrices of order 3×3

$$A=\begin{pmatrix}2 & -1 & 3\\ 0 & 4 & 6\\ -6 & 10 & -5\end{pmatrix},\quad B=\begin{pmatrix}4 & 7 & -8\\ 9 & 3 & 5\\ 1 & -1 & 2\end{pmatrix}$$

Since the given matrices have same orders. Therefore, these matrices can be added and their sum is given by

$$A+B=\begin{pmatrix}2+4 & -1+7 & 3+(-8)\\ 0+9 & 4+3 & 6+5\\ -6+1 & 10+(-1) & -5+2\end{pmatrix}=\begin{pmatrix}6 & 6 & -5\\ 9 & 7 & 11\\ -5 & 9 & -3\end{pmatrix}$$

Example 3 Write the following single column matrix as the sum of three column vectors

$$\begin{pmatrix}3t^2-2e^t\\ t^2+7t\\ 5t\end{pmatrix}$$

Solution

$$\begin{pmatrix}3t^2-2e^t\\ t^2+7t\\ 5t\end{pmatrix}=\begin{pmatrix}3t^2\\ t^2\\ 0\end{pmatrix}+\begin{pmatrix}0\\ 7t\\ 5t\end{pmatrix}+\begin{pmatrix}-2e^t\\ 0\\ 0\end{pmatrix}=\begin{pmatrix}3\\ 1\\ 0\end{pmatrix}t^2+\begin{pmatrix}0\\ 7\\ 5\end{pmatrix}t+\begin{pmatrix}-2\\ 0\\ 0\end{pmatrix}e^t$$

38.9 Difference of Matrices

The difference of two matrices A and B of same order $m\times n$ is defined to be the matrix

$$A-B=A+(-B)$$

The matrix $-B$ is obtained by multiplying the matrix B with -1. So that

$$-B=(-1)B$$

Scanned with CamScanner

$$X = \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \\ \vdots \\ b_{n1} \end{pmatrix} = [b_{i1}]_{n\times 1}$$

A column matrix is also called a column vector or simply a vector.

38.7 Multiple of matrices

A multiple of a matrix A is defined to be

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{bmatrix} = [ka_{ij}]_{m\times n}$$

Where k is a constant or it is a function. Notice that the product kA is same as product Ak. Therefore, we can write

$kA = Ak$ or $AK = KA$

Example 1

(a) $$5\begin{bmatrix} 2 & -3 \\ 4 & -1 \\ 1/5 & 6 \end{bmatrix} = \begin{bmatrix} 10 & -15 \\ 20 & -5 \\ 1 & 30 \end{bmatrix}$$

(b) $$e^t \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} e^t \\ -2e^t \\ 4e^t \end{bmatrix}$$

Since we know that $kA = Ak$. Therefore, we can write $e^{-3t}\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} e^{-3t}$

Scanned with CamScanner

For $r_1 = \frac{1}{2}$, we can divide by $k + \frac{3}{2}$ in the above equation to obtain

$$c_{k+1} = \frac{-c_k}{2(k+1)},$$

$$c_1 = \frac{-c_0}{2.1}$$

$$c_2 = \frac{-c_1}{2.2} = \frac{c_0}{2^2.2!}$$

$$c_3 = \frac{-c_2}{2.3} = \frac{-c_0}{2^3.3!}$$

$$\vdots$$

In general $$c_n = \frac{(-1)^n c_0}{2^n n!}, \quad n = 1, 2, 3, \ldots$$

Thus we have $y_1 = c_0 x^{\frac{1}{2}} \left[1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2^n n!} x^n\right]$, which converges for $x \geq 0$. As given, the series is not meaningful for $x < 0$ because of the presence of $x^{\frac{1}{2}}$.

Now for $r_2 = 0$, (1) becomes

$$c_{k+1} = \frac{-c_k}{2k+1}$$

$$c_1 = \frac{-c_0}{1}$$

$$c_2 = \frac{-c_1}{3} = \frac{c_0}{1.3}$$

$$c_3 = \frac{-c_2}{5} = \frac{-c_0}{1.3.5}$$

$$c_4 = \frac{-c_3}{7} = \frac{c_0}{1.3.5.7}$$

$$\vdots$$

In general $$c_n = \frac{(-1)^n c_0}{1.3.5.7\ldots(2n-1)}, \quad n = 1, 2, 3, \ldots$$

Thus second solution is

Scanned with CamScanner

38.13 Associative Law

The matrix multiplication is associative. This means that if A, B and C are $m \times p$, $p \times r$ and $r \times n$ matrices, then $A(BC) = (AB)C$. The result is a $m \times n$ matrix.

38.14 Distributive Law

If B and C are matrices of order $r \times n$ and A is a matrix of order $m \times r$, then the distributive law states that $A(B+C) = AB + AC$

Furthermore, if the product $(A+B)C$ is defined, then $(A+B)C = AC + BC$

38.15 Determinant of a Matrix

Associated with every square matrix A of constants, there is a number called the determinant of the matrix, which is denoted by $\det(A)$ or $|A|$

Example 6 Find the determinant of the following matrix

$$A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$$

Solution

The determinant of the matrix A is given by

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

We expand the $\det(A)$ by cofactors of the first row, we obtain

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix} = 3\begin{vmatrix} 5 & 1 \\ 2 & 4 \end{vmatrix} - 6\begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} + 2\begin{vmatrix} 2 & 5 \\ -1 & 2 \end{vmatrix}$$

or

$$\det(A) = 3(20-2) - 6(8+1) + 2(4+5) = 18$$

38.16 Transpose of a Matrix

The transpose of a $m \times n$ matrix A is obtained by interchanging rows and columns of the matrix and is denoted by A^{tr}. In other words, rows of A become the columns of A^{tr}. If

Scanned with CamScanner

(a)
$$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ 1 & -7 & 9 \end{pmatrix}\begin{pmatrix} -3 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 2\cdot(-3)+(-1)\cdot 6+3\cdot 4 \\ 0\cdot(-3)+4\cdot 6+5\cdot 6 \\ 1\cdot(-3)+(-7)\cdot 6+9\cdot 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 44 \\ -9 \end{pmatrix}$$

(b)
$$\begin{pmatrix} -4 & 2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4x+2y \\ 3x+8y \end{pmatrix}$$

38.11 Multiplicative Identity

For a given positive integer n, the $n \times n$ matrix

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

is called the multiplicative identity matrix. If A is a matrix of order $n \times n$, then it can verified that

$$I \cdot A = A \cdot I = A$$

Also, it is readily verified that if X is any $n \times 1$ column matrix, then $I \cdot X = X$

38.12 Zero Matrix

A matrix consisting of all zero entries is called a zero matrix or null matrix and is denote by O. For example

$$O = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and so on. If A and O are $m \times n$ matrices, then

$$A + O = O + A = A$$

Scanned with CamScanner

(b) $$A=\begin{pmatrix}5 & 8\\ 1 & 0\\ 2 & 7\end{pmatrix}, \quad B=\begin{pmatrix}-4 & -3\\ 2 & 0\end{pmatrix}$$

Solution

(a) The matrices A and B are square matrices of order 2. Therefore, both of the products AB and BA are possible.

$$AB=\begin{pmatrix}4 & 7\\ 3 & 5\end{pmatrix}\begin{pmatrix}9 & -2\\ 6 & 8\end{pmatrix}=\begin{pmatrix}4\cdot 9+7\cdot 6 & 4\cdot(-2)+7\cdot 8\\ 3\cdot 9+5\cdot 6 & 3\cdot(-2)+5\cdot 8\end{pmatrix}=\begin{pmatrix}78 & 48\\ 57 & 34\end{pmatrix}$$

Similarly
$$BA=\begin{pmatrix}9 & -2\\ 6 & 8\end{pmatrix}\begin{pmatrix}4 & 7\\ 3 & 5\end{pmatrix}=\begin{pmatrix}9\cdot 4+(-2)\cdot 3 & 9\cdot 7+(-2)\cdot 5\\ 6\cdot 4+8\cdot 3 & 6\cdot 7+8\cdot 5\end{pmatrix}=\begin{pmatrix}30 & 53\\ 48 & 82\end{pmatrix}$$

(b) The product AB is possible as the number of columns in the matrix A and the number of rows in B is 2. However, the product BA is not possible because the number of rows in the matrix B and the number of rows in A is not same.

Columns

$$AB=\begin{pmatrix}5\cdot(-4)+8\cdot 2 & 5\cdot(-3)+8\cdot 0\\ 1\cdot(-4)+0\cdot 2 & 1\cdot(-3)+0\cdot 0\\ 2\cdot(-4)+7\cdot 2 & 2\cdot(-3)+7\cdot 0\end{pmatrix}=\begin{pmatrix}-4 & -15\\ -4 & -3\\ 6 & -6\end{pmatrix}$$

Note that

In general, matrix multiplication is not commutative. This means that $AB \neq BA$. For example, we observe in part (a) of the previous example

$$AB=\begin{pmatrix}78 & 48\\ 57 & 34\end{pmatrix}, \quad BA=\begin{pmatrix}30 & 53\\ 48 & 82\end{pmatrix}$$

Clearly $AB \neq BA$.. Similarly in part (b) of the example, we have

$$AB=\begin{pmatrix}-4 & -15\\ -4 & -3\\ 6 & -6\end{pmatrix}$$

However, the product BA is not possible.

Example 5

Scanned with CamScanner

$$y_1 = c_0 x^{\frac{2}{3}}\left[1+\sum_{n=1}^{\infty}\frac{1}{n!5.8.11.14...(3n+2)}x^n\right] \qquad (3)$$

$$y_2 = c_0 x^{0}\left[1+\sum_{n=1}^{\infty}\frac{1}{n!1.4.7...(3n-2)}x^n\right]. \qquad (4)$$

By the ratio test it can be demonstrated that both (3) and (4) converge for all finite of x. Also it should be clear from the form of (3) and (4) that neither series is a multiple of the other and therefore, $y_1(x)$ and $y_2(x)$ are linearly independent on axis. Hence by the superposition principle

$$y = C_1 y_1(x) + C_2 y_2(x) = C_1\left[x^{\frac{2}{3}}+\sum_{n=1}^{\infty}\frac{1}{n!5.8.11.14...(3n+2)}x^{n+\frac{2}{3}}\right]$$

$$+C_2\left[1+\sum_{n=1}^{\infty}\frac{1}{n!1.4.7...(3n-2)}x^n\right], \quad |x|<\infty$$

is an other solution of the differential equation. On any interval not containing the origin, this combination represents the general solution of the differential equation

Note: The method of Frobenius may not always provide 2 solutions.

Example: The differential equation $xy''+3y'-y=0$ has regular singular point at

We try a solution of the form $y=\sum_{n=0}^{\infty}c_n x^{n+r}$

Therefore $y'=\sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1}$ and $y''=\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-2}$.

so that $xy''+3y'-y = x^r\left[r(r+2)c_0x^{-1}+\sum_{k=0}^{\infty}[(k+r+1)(k+r+3)c_{k+1}-c_k]x^k\right]$

so that the indicial equation and exponent are $r(r+2)=0$ and $r_1=0$, respectively.

Since $(k+r+1)(k+r+3)c_{k+1}-c_k=0$, $\quad k=0,1,2,\ldots \qquad (1)$

it follows that when $r_1=0$,

$$c_{k+1}=\frac{c_k}{(k+1)(k+3)},$$

Scanned with CamScanner

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \cdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Then

$$A^{tr} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

Since order of the matrix A is $m \times n$, the order of the transpose matrix A^{tr} is $n \times m$

Example 7

(a) The transpose of matrix

$$A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$$

is

$$A^{tr} = \begin{pmatrix} 3 & 2 & -1 \\ 6 & 5 & 2 \\ 2 & 1 & 4 \end{pmatrix}$$

(b) If X denotes the matrix

$$X = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$$

Then

$$X^{tr} = \begin{bmatrix} 5 & 0 & 3 \end{bmatrix}$$

© Copyright Virtual University

Scanned with CamScanner

2. For a 3×3 non-singular matrix

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$C_{11}=\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, C_{12}=-\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \quad C_{13}=\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

and so on. Therefore, inverse of the matrix A is given by

$$A^{-1}=\frac{1}{\det A}\begin{pmatrix} C_{11} & C_{21} & a_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}.$$

Example 8 short

Find, if possible, the multiplicative inverse for the matrix

$$A=\begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}.$$

Solution:

The matrix A is non-singular because

$$\det(A)=\begin{vmatrix} 1 & 4 \\ 2 & 10 \end{vmatrix}=10-8=2$$

Therefore, A^{-1} exists and is given by

$$A^{-1}=\frac{1}{2}\begin{pmatrix} 10 & -4 \\ -2 & 1 \end{pmatrix}=\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}$$

Check

© Copyright Virtual

Scanned with CamScanner

$$C_{21} = -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -2, \quad C_{22} = \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} = 2, \quad C_{23} = -\begin{vmatrix} 2 & 2 \\ 3 & 0 \end{vmatrix} = 6$$

$$C_{31} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \quad C_{32} = -\begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} = -2, \quad C_{33} = \begin{vmatrix} 2 & 2 \\ -2 & 1 \end{vmatrix} = 6$$

Hence $$A^{-1} = \frac{1}{12}\begin{pmatrix} 1 & -2 & 2 \\ 5 & 2 & -2 \\ -3 & 6 & 6 \end{pmatrix} = \begin{pmatrix} 1/12 & -1/6 & 1/6 \\ 5/12 & 1/6 & -1/6 \\ -1/4 & 1/2 & 1/2 \end{pmatrix}$$

Please verify that $A \cdot A^{-1} = A^{-1} \cdot A = I$

38.19 Derivative of a Matrix of functions

Suppose that

$$A(t) = \left[a_{ij}(t)\right]_{m\times n}$$

is a matrix whose entries are functions those are differentiable on a common interval then derivative of the matrix $A(t)$ is a matrix whose entries are derivatives of the corresponding entries of the matrix $A(t)$. Thus

$$\frac{dA}{dt} = \left[\frac{da_{ij}}{dt}\right]_{m\times n}$$

The derivative of a matrix is also denoted by $A'(t)$.

38.20 Integral of a Matrix of Functions

Suppose that $A(t) = \left(a_{ij}(t)\right)_{m\times n}$ is a matrix whose entries are functions those are continuous on a common interval containing t, then integral of the matrix $A(t)$ is a matrix whose entries are integrals of the corresponding entries of the matrix $A(t)$. Thus

$$\int_{t_0}^{t} A(s)ds = \left(\int_{t_0}^{t} a_{ij}(s)ds\right)_{m\times n}$$

Example 11

Find the derivative and the integral of the following matrix

$$X(t) = \begin{pmatrix} \sin 2t \\ e^{3t} \\ 8t - 1 \end{pmatrix}$$

Solution:

Scanned with CamScanner

$$AA^{-1}=\begin{pmatrix}1 & 4\\ 2 & 10\end{pmatrix}\begin{pmatrix}5 & -2\\ -1 & 1/2\end{pmatrix}=\begin{pmatrix}5-4 & -2+2\\ 10-10 & -4+5\end{pmatrix}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=I$$

$$AA^{-1}=\begin{pmatrix}5 & -2\\ -1 & 1/2\end{pmatrix}\begin{pmatrix}1 & 4\\ 2 & 10\end{pmatrix}=\begin{pmatrix}5-4 & 20-20\\ -1+1 & -4+5\end{pmatrix}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=I$$

Example 9 Short

Find, if possible, the multiplicative inverse of the following matrix

$$A=\begin{pmatrix}2 & 2\\ 3 & 3\end{pmatrix}$$

Solution:

The matrix is singular because

$$\det(A)=\begin{vmatrix}2 & 2\\ 3 & 3\end{vmatrix}=2\cdot 3-2\cdot 3=0$$

Therefore, the multiplicative inverse A^{-1} of the matrix does not exist.

Example 10 Short important

Find the multiplicative inverse for the following matrix

$$A=\begin{pmatrix}2 & 2 & 0\\ -2 & 1 & 1\\ 3 & 0 & 1\end{pmatrix}.$$

Solution:

Since $$\det(A)=\begin{vmatrix}2 & 2 & 0\\ -2 & 1 & 1\\ 3 & 0 & 1\end{vmatrix}=2(1-0)-2(-2-3)+0(0-3)=12\neq 0$$

Therefore, the given matrix is non singular. So that, the multiplicative inverse A^{-1} of the matrix A exists. The cofactors corresponding to the entries in each row are

$$C_{11}=\begin{vmatrix}1 & 1\\ 0 & 1\end{vmatrix}=1, \qquad C_{12}=-\begin{vmatrix}-2 & 1\\ 3 & 1\end{vmatrix}=5, \qquad C_{13}=\begin{vmatrix}-2 & 1\\ 3 & 0\end{vmatrix}=-3$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

38.17 Multiplicative Inverse of a Matrix

Suppose that A is a square matrix of order $n \times n$. If there exists an $n \times n$ matrix B such that

$$AB = BA = I$$

Then B is said to be the multiplicative inverse of the matrix A and is denoted by $B = A^{-1}$.

38.18 Non-Singular Matrices

A square matrix A of order $n \times n$ is said to be a non-singular matrix if

$$\det(A) \neq 0$$

Otherwise the square matrix A is said to be singular. Thus for a singular A we must have

$$\det(A) = 0$$

Theorem If A is a square matrix of order $n \times n$ then the matrix has a multiplicative inverse A^{-1} if and only if the matrix A is non-singular. Quiz

Theorem Let A be a non singular matrix of order $n \times n$ and let C_{ij} denote the cofactor (signed minor) of the corresponding entry a_{ij} in the matrix A i.e. Quiz

$$C_{ij} = (-1)^{i+j} M_{ij}$$

M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting the ith row and jth column from A. Then inverse of the matrix A is given by

$$A^{-1} = \frac{1}{\det(A)} (C_{ij})^{tr} \quad \text{Quiz}$$

12:40

Further Explanation

1. For further reference we take $n = 2$ so that A is a 2×2 non-singular matrix given by

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Therefore $C_{11} = a_{22}$, $C_{12} = -a_{21}$, $C_{21} = -a_{12}$ and $C_{22} = a_{11}$. So that

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{pmatrix}^{tr} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Scanned with CamScanner

39 The Eigenvalue problem

39.1 Eigenvalues and Eigenvectors

Let A be a $n \times n$ matrix. A number λ is said to be an eigenvalue of A if there exists a nonzero solution vector K of the system of linear differential equations

$$AK = \lambda K$$

The solution vector K is said to be an eigenvector corresponding to the eigenvalue λ. Using properties of matrix algebra, we can write the above equation in the following alternative form

$$(A - \lambda I)K = 0$$

where I is the identity matrix.

If we let $$K = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ \vdots \\ k_n \end{pmatrix}$$

Then the above system is same as the following system of linear algebraic equations

$$\begin{aligned} (a_{11} - \lambda)k_1 + a_{12}k_2 + \cdots + a_{1n}k_n &= 0 \\ a_{21}k_1 + (a_{22} - \lambda)k_2 + \cdots + a_{2n}k_n &= 0 \\ \vdots \qquad\qquad \vdots \qquad\qquad &\vdots \\ a_{n1}k_n + a_{n2}k_2 + \cdots + (a_{nn} - \lambda)k_n &= 0 \end{aligned}$$

Clearly, an obvious solution of this system is the trivial solution

$$k_1 = k_2 = \ldots = k_n = 0$$

However, we are seeking only a non-trivial solution of the system.

39.2 The Non-trivial solution

The non-trivial solution of the system exists only when

$$\det(A - \lambda I) = 0$$

Quiz

Ans = (true)

very important

This equation is called the characteristic equation of the matrix A. Thus the Eigenvalues of the matrix A are given by the roots of the characteristic equation. To find an eigenvector corresponding to an eigenvalue λ we simply solve the system of linear algebraic equations

$$\det(A - \lambda I)K = 0$$

© Copyright

Scanned with CamScanner

$$A_b = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{211} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{bmatrix}$$

If B denotes the column matrix of the b_i, $\forall i = 1, 2, \ldots, n$ then the augmented matrix the above mentioned system of linear algebraic equations can be written as $(A \mid B)$.

38.22 Elementary Row Operations

Quiz

The elementary row operations consist of the following three operations

- Multiply a row by a non-zero constant.
- Interchange any row with another row.
- Add a non-zero constant multiple of one row to another row.

These row operations on the augmented matrix of a system are equivalent to, multiplyin an equation by a non-zero constant, interchanging position of any two equations of th system and adding a constant multiple of an equation to another equation.

38.23 The Gaussian and Gauss-Jordon Methods

Quiz + Short

In the Gaussian Elimination method we carry out a succession of elementary ro operations on the augmented matrix of the system of linear equations to be solved until i is transformed into row-echelon form, a matrix that has the following structure:

- The first non-zero entry in a non-zero row is 1.
- In consecutive nonzero rows the first entry 1 in the lower row appears to the right of the first 1 in the higher row.
- Rows consisting of all 0's are at the bottom of the matrix.

In the Gauss-Jordan method the row operations are continued until the augmented matrix is transformed into the reduced row-echelon form. A reduced row-echelon matrix has the structure similar to row-echelon, but with an additional property.

- The first non-zero entry in a non-zero row is 1.
- In consecutive nonzero rows the first entry 1 in the lower row appears to the right of the first 1 in the higher row.
- Rows consisting of all 0's are at the bottom of the matrix.
- A column containing a first entry 1 has 0's everywhere else.

Example 1

(a) The following two matrices are in row-echelon form.

Scanned with CamScanner

The derivative and integral of the given matrix are, respectively, given by

$$X'(t) = \begin{pmatrix} \frac{d}{dt}(\sin 2t) \\ \frac{d}{dt}(e^{3t}) \\ \frac{d}{dt}(8t-1) \end{pmatrix} = \begin{pmatrix} 2\cos 2t \\ 3e^{3t} \\ 8 \end{pmatrix}$$

$$\int_0^t X(s)ds = \begin{pmatrix} \int_0^t \sin 2t ds \\ \int_0^t e^{3t} ds \\ \int_0^t 8t-1 ds \end{pmatrix} = \begin{pmatrix} -1/2\cos 2t + 1/2 \\ 1/3e^{3t} - 1/3 \\ 4t^2 - t \end{pmatrix}$$

short + Quiz

38.21 Augmented Matrix

Consider an algebraic system of n linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad\quad \vdots \quad\quad\quad \vdots \quad\quad \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Suppose that A denotes the coefficient matrix in the above algebraic system, then

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Quiz

It is well known that Cramer's rule can be used to solve the system, whenever $\det(A) \neq 0$. However, it is also well known that a Herculean effort is required to solve the system if $n > 3$. Thus for larger systems the Gaussian and Gauss-Jordon elimination methods are preferred and in these methods we apply elementary row operations on augmented matrix.

Quiz

The augmented matrix of the system of linear equations is the following $n \times (n+1)$ matrix

© Copyright Virtual University of Pakistan

Scanned with CamScanner

This system of equations can be solved by applying the Gauss-Jordan elimination to the augmented matrix

$$(A-\lambda I \;|0)$$

Verify that the following column vector is an eigenvector

$$K=\begin{pmatrix}1\\-1\\1\end{pmatrix}$$

is an eigenvector of the following 3×3 matrix

$$A=\begin{pmatrix}0 & -1 & -3\\2 & 3 & 3\\-2 & 1 & 1\end{pmatrix}$$

Solution:

By carrying out the multiplication AK, we see that

$$AK=\begin{pmatrix}0 & -1 & -3\\2 & 3 & 3\\-2 & 1 & 1\end{pmatrix}\begin{pmatrix}1\\-1\\1\end{pmatrix}=(-2)\begin{pmatrix}-2\\2\\1\end{pmatrix}=(-2)K$$

Hence the number $\lambda=-2$ is an eigenvalue of the given matrix A.

Example 5

Find the eigenvalues and eigenvectors of

only

$$A=\begin{pmatrix}1 & 2 & 1\\6 & -1 & 0\\-1 & -2 & -1\end{pmatrix}$$

Solution:

Eigenvalues Short

The characteristic equation of the matrix A is

$$\det(A-\lambda I)=\begin{vmatrix}1-\lambda & 2 & 1\\6 & -1-\lambda & 0\\-1 & -2 & -1-\lambda\end{vmatrix}=0$$

Expanding with respect to 3rd column

Scanned with CamScanner

$$\left(\begin{array}{ccc|c} 1 & 5 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccccc|c} 0 & 0 & 1 & -6 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

Please verify that the three conditions of the structure of the echelon form are satisfied.

(b) The following two matrices are in reduced row-echelon form.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccccc|c} 0 & 0 & 1 & -6 & 0 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

Please notice that all remaining entries in the columns containing a leading entry 1 are 0.

Notation

To keep track of the row operations on an augmented matrix, we utilized the following notation:

very important

Symbol	Meaning
R_{ij}	Interchange the rows i and j.
cR_i	Multiply the ith row by a nonzero constant c.
$cR_i + R_j$	Multiply the ith row by c and then add to the jth row

Column

Example 2

Solve the following system of linear algebraic equations by the (a) Gaussian elimination and (b) Gauss-Jordan elimination

$$\begin{aligned} 2x_1 + 6x_2 + x_3 &= 7 \\ x_1 + 2x_1 - x_3 &= -1 \\ 5x_1 + 7x_2 - 4x_3 &= 9 \end{aligned}$$

Solution

(a) The augmented matrix of the system is

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$K_1 = \begin{pmatrix} 1 \\ \frac{6}{5} \\ -13 \end{pmatrix}$$

For $\lambda_2 = -4$, we have

$$(A+4I\ |0) = \left(\begin{array}{ccc|c} 5 & 2 & 1 & 0 \\ 6 & 3 & 0 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right)$$

By $(-1)R_3,\ R_{32}$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 6 & 3 & 0 & 0 \\ 5 & 2 & 1 & 0 \end{array}\right)$$

By $-6R_1 + R_2, -5R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -9 & 18 & 0 \\ 0 & -8 & 16 & 0 \end{array}\right)$$

By $-\frac{1}{9}R_2,\ -\frac{1}{8}R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right)$$

By $-2R_2 + R_1,\ -R_2 + R_3$ $\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

Hence we obtain the following two equations involving k_1, k_2 and k_3.

$$k_1 = -k_3,\ k_2 = 2k_3$$

Choosing $k_3 = 1$, we have $k_1 = -1,\ k_2 = 2$. Hence we have an eigenvector corresponding to the eigenvalue $\lambda_2 = -4$, $K_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Finally, for $\lambda_3 = 3$, we have $(A-3I|0) = \begin{pmatrix} -2 & 2 & 1 & 0 \\ 6 & -4 & 0 & 0 \\ -1 & -2 & -4 & 0 \end{pmatrix}$

By using the Gauss Jordon elimination as used for other values, we obtain (verify)

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 3/2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So that we obtain the equations $k_1 = -k_3, \quad k_2 = (-3/2)k_3$

The choice $k_3 = -2$ leads to $k_1 = 2, \quad k_2 = 3$. Hence, we have the following eigenvector

$$K_3 = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$$

Note that: Quiz.

The component k_3 could be chosen as any nonzero number. Therefore, a nonzero constant multiple of an eigenvector is also an eigenvector.

Example 6 Important Short + Quiz

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$

Solution:

From the characteristic equation of the given matrix is

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 4 \\ -1 & 7-\lambda \end{vmatrix} = 0$$

or

$$(3-\lambda)(7-\lambda)+4 = 0 \Rightarrow (\lambda-5)^2 = 0$$

Therefore, the characteristic equation has repeated real roots. Thus the matrix has an eigenvalue of multiplicity two. $\lambda_1 = \lambda_2 = 5$

In the case of a 2×2 matrix there is no need to use Gauss-Jordan elimination. To find the eigenvector(s) corresponding to $\lambda_1 = 5$ we resort to the system of linear equations

$$(A-5I)K = 0$$

or in its equivalent form

$$-2k_1 + 4k_2 = 0$$
$$-k_1 + 2k_2 = 0$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$\Rightarrow \begin{vmatrix} 6 & -1-\lambda \\ -1 & -2 \end{vmatrix} - 0 + (-1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 6 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda - 13 + (-\lambda - 1)(\lambda^2 - 13) = 0$$

$$\Rightarrow -\lambda(\lambda + 4)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 0, -4, 3$$

Eigenvectors

For $\lambda_1 = 0$ we have

$$(A - 0|0) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 6 & -1 & 0 & 0 \\ -1 & -2 & -1 & 0 \end{array}\right)$$

By $-6R_1 + R_2,\ R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -13 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

By $-\frac{1}{13}R_2$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 6/13 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

By $-2R_2 + R_1$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1/13 & 0 \\ 0 & 1 & 6/13 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus we have the following equations in k_1, k_2 and k_3. The number k_3 can be chos arbitrarily

$$k_1 = -(1/13)k_3,\ k_2 = -(6/13)k_3$$

Choosing $k_3 = -13$, we get $k_1 = 1$ and $k_2 = 6$. Hence, the eigenvector corresponding $\lambda_1 =$ is

Scanned with CamScanner

It is apparent from this system that $k_1 = 2k_2$.

Thus if we choose $k_2 = 1$, we find the single eigenvector $K_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Example 7

Find the eigenvalues only and eigenvectors of

$$A = \begin{pmatrix} 9 & 1 & 1 \\ 1 & 9 & 1 \\ 1 & 1 & 9 \end{pmatrix}$$

Solution Short

The characteristic equation of the given matrix is

$$\det(A - \lambda I) = \begin{vmatrix} 9-\lambda & 1 & 1 \\ 1 & 9-\lambda & 1 \\ 1 & 1 & 9-\lambda \end{vmatrix} = 0$$

Or $(\lambda - 11)(\lambda - 8)^2 = 0 \Rightarrow \lambda = 11,\ 8,\ 8$

Thus the eigenvalues of the matrix are $\lambda_1 = 11, \lambda_2 = \lambda_3 = 8$

For $\lambda_1 = 11$, we have $(A - 11I \mid 0) = \left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right)$

The Gauss-Jordan elimination gives $\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

$$K_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Hence, $k_1 = k_3$, $k_2 = k_3$. If $k_3 = 1$, then

Now for $\lambda_2 = 8$ we have, $(A - 8I \mid 0) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right)$

Again the Gauss-Jordon elimination gives $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$

Scanned with CamScanner

8. $\begin{pmatrix} 2 & -1 & 0 \\ 5 & 2 & 4 \\ 0 & 1 & 2 \end{pmatrix}$

Important for short and objective

40 Matrices and Systems of Linear First-Order Equations

40.1 Matrix form of a system

Consider the following system of linear first-order differential equations

$$\begin{aligned}
\frac{dx_1}{dt} &= a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t) \\
\frac{dx_2}{dt} &= a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t) \\
&\vdots \\
\frac{dx_n}{dt} &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t)
\end{aligned}$$

Suppose that X, $A(t)$ and $F(t)$, respectively, denote the following matrices

$$X = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad A(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix}, \quad F(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

Then the system of differential equations can be written as

$$\frac{d}{dt}\begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix}\begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} + \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Therefore, $k_1 + k_2 + k_3 = 0$

We are free to select two of the variables arbitrarily. Choosing, on the one hand, $k_2 = 1, k_3 = 0$ and, on the other, $k_2 = 0, k_3 = 1$, we obtain two linearly independent eigenvectors corresponding to a single eigenvalue $K_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $K_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

Note that

Thus we note that when a $n \times n$ matrix A possesses n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, a set of n linearly independent eigenvectors $K_1, K_2, \ldots, K_n$ can be found. However, when the characteristic equation has repeated roots, it may not be possible to find n linearly independent eigenvectors of the matrix.

39.3 Exercise

Find the eigenvalues and eigenvectors of the given matrix.

1. $\begin{pmatrix} -1 & 2 \\ -7 & 8 \end{pmatrix}$
2. $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$
3. $\begin{pmatrix} -8 & -1 \\ 16 & 0 \end{pmatrix}$
4. $\begin{pmatrix} 5 & -1 & 0 \\ 0 & -5 & 9 \\ 5 & -1 & 0 \end{pmatrix}$
5. $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$
6. $\begin{pmatrix} 0 & 4 & 0 \\ -1 & -4 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

Show that the given matrix has complex eigenvalues.

7. $\begin{pmatrix} -1 & 2 \\ -5 & 1 \end{pmatrix}$

© Cam...

Scanned with CamScanner

or simply

$$\frac{dX}{dt} = A(t)X + F(t)$$

If the system of differential equations is homogenous, then $F(t) = 0$ and we can write

$$\frac{dX}{dt} = A(t)X$$

Both the non-homogeneous and the homogeneous systems can also be written as

$$X' = AX + F, \quad X' = AX$$

Example 1 for short or Quiz

Write the following non-homogeneous system of differential equations in the matrix form

$$\frac{dx}{dt} = -2x + 5y + e^t - 2t$$

$$\frac{dy}{dt} = 4x - 3y + 10t$$

Solution:

If we suppose that

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

Then, the given non-homogeneous differential equations can be written as

$$\frac{dX}{dt} = \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} e^t - 2t \\ 10t \end{pmatrix}$$

or

$$X' = \begin{pmatrix} -2 & 5 \\ 4 & -3 \end{pmatrix} X + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} -2 \\ 10 \end{pmatrix} t$$

Solution Vector

Consider a homogeneous system of differential equations

$$\frac{dX}{dt} = AX$$

A solution vector on an interval I of the homogeneous system is any column matrix

©Copyright Virtual University of Pakistan

Scanned with CamScanner

and

$$AX_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ e^{4t} \end{pmatrix} = \begin{pmatrix} 0 \\ e^{4t} \end{pmatrix} = X_2$$

or

$$AX_2 = \begin{pmatrix} 1 \\ e^{4t} \end{pmatrix} = X_2$$

Thus, the vectors X_1 and X_2 satisfy the homogeneous linear system

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} X$$

Hence, the given vectors are solutions of the given homogeneous system of differential equations.

Note that

Much of the theory of the systems of a linear first order differential equations is similar to that of the linear nth order differential equations.

10.2 Initial -Value Problem Next class

Let I_0 denote any point in some interval denoted by I and

$$X(t_0) = \begin{pmatrix} x_1(t_0) \\ x_2(t_0) \\ \vdots \\ x_n(t_0) \end{pmatrix}, \quad X_0 = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix}$$

$k_i (i = 1, 2, \ldots, n)$ are given constants. Then the problem of solving the system of differential equations

$$\frac{dX}{dt} = A(t)X + F(t)$$

Subject to the initial conditions

$$X(t_0) = X_0$$

is called an initial value problem on the interval I.

Scanned with CamScanner

$$\frac{dX}{dt} = c_1 \frac{dX}{dt} = c_1 \begin{pmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t + \sin t \end{pmatrix}$$

Also

$$AX = c_1 \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

$$AX = c_1 \begin{pmatrix} -\sin t \\ \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t + \sin t \end{pmatrix}$$

Thus, we have verified that:

$$\frac{dX}{dt} = AX$$

Hence the vector $c_1 X_1$ is also a solution vector of the homogeneous system of differential equations.

Example 4 For short and Long

Consider the following system considered in the previous example 4

$$X' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} X$$

We know from the previous example that the vector X_1 is a solution of the system

$$X_1 = \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

40.3 Theorem: Existence of a unique Solution for short and Quiz

Suppose that the entries of the matrices $A(t)$ and $F(t)$ in the system of differential equations $\frac{dX}{dt} = A(t)X + F(t)$ being considered in the above mentioned initial value problem, are continuous functions on a common interval I that contains the point t_0. Then there exist a unique solution of the initial–value problem on the interval I.

40.4 Superposition Principle for short and Quiz

Suppose that $X_1, X_2, \ldots, X_n$ be a set of solution vectors of the homogeneous system

$$\frac{dX}{dt} = A(t)X$$

on an interval I. Then the principle of superposition states that linear combination

$$X = c_1X_1 + c_2X_2 + \cdots + c_kX_k$$

$c_i, i = 1, 2, \ldots, k$ being arbitrary constants, is also a solution of the system on the same interval I.

Note that

An immediate consequence of the principle of superposition is that a constant multiple of any solution vector of a homogenous system of first order differential equation is also a solution of the system.

Example 3 for short

Consider the following homogeneous system of differential equations

$$X' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} X$$

Also consider a solution vector X_1 of the system that is given by

$$X_1 = \begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix}$$

For any constant c_1 the vector $X = c_1X_1$ is also a solution of the homogeneous system.

To verify this we differentiae the vector X with respect to t

© Copyright Virtual University of Pakistan

Scanned with CamScanner

If $$X_2 = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

Then $$X'_2 = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

and $$AX_2 = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

Therefore

$$AX_2 = X_2'$$

Hence the vector X_2 is a solution vector of the homogeneous system. We can verify that the following vector is also a solution of the homogeneous system.

$$X = c_1X_1 + c_2X_2$$

or $$X = c_1\begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix} + c_2\begin{pmatrix} 0 \\ e^t \\ 0 \end{pmatrix}$$

40.5 Linear Dependence of Solution Vectors

Let $X_1, X_2, X_3, \ldots, X_k$ be a set of solution vectors, on an interval I, of the homogenous system of differential equations

$$\frac{dX}{dt} = AX$$

We say that the set is linearly dependent on I if there exist constants $c_1, c_2, c_3 \ldots, c_k$ not all zero such that

$$X(t) = c_1X_1(t) + c_2X_2(t) + \cdots + c_kX_k(t) = 0, \quad \forall\, t \in I$$

Note that

- Any two solution vectors X_1 and X_2 are linearly dependent if and only if one of the two vectors is a constant multiple of the other.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$c_1X_1 + c_2X_2 = 0$$

$$c_1\begin{pmatrix}3\\1\end{pmatrix}e^{t} + c_2\begin{pmatrix}1\\1\end{pmatrix}e^{-t} = \begin{pmatrix}0\\0\end{pmatrix}$$

This means that

$$3c_1e^{t} + c_2e^{-t} = 0$$

$$c_1e^{t} + c_2e^{-t} = 0$$

The only solution of these equations for the arbitrary constants c_1 and c_2 is

$$c_1 = c_2 = 0$$

Hence, the solution vectors X_1 and X_2 are linearly independent on $(-\infty, \infty)$.

Example 6 For short and Long

Again consider the same homogeneous system as considered in the previous example

$$X' = \begin{pmatrix}2 & -3\\1 & -2\end{pmatrix}X$$

We have already seen that the vectors X_1, X_2 i.e.

$$X_1 = \begin{pmatrix}3e^{t}\\e^{t}\end{pmatrix}, \quad X_2 = \begin{pmatrix}e^{-t}\\e^{-t}\end{pmatrix}$$

are solutions of the homogeneous system. We can verify that the following vector X_3

$$X_3 = \begin{pmatrix}e^{t} + \cosh t\\ \cosh t\end{pmatrix}$$

is also a solution of the homogeneous system However, the set of solutions that consists of X_1, X_2 and X_3 is linearly dependent because X_3 is a linear combination of the other two vectors

$$X_3 = \frac{1}{2}X_1 + \frac{1}{2}X_2$$

40.7 Exercise

Write the given system in matrix form.

© Copyright Virtual University of Pakistan

Scanned with CamScanner

- For $k > 2$ if the set of k solution vectors is linearly dependent then we can express at least one of the solution vectors as a linear combination of the remaining vectors.

40.6 Linear Independence of Solution Vectors

Suppose that $X_1, X_2, \ldots, X_k$ is a set of solution vectors, on an interval I, of the homogenous system of differential equations

$$\frac{dX}{dt} = AX$$

Then the set of solution vectors is said to be linearly independent if it is not linearly dependent on the interval I. This means that

$$X(t) = c_1X_1(t) + c_2X_2(t) + \cdots + c_kX_k(t) = 0$$

only when each $c_i = 0$.

Example 5

Consider the following two column vectors

$$X_1 = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}, \quad X_2 = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$$

Since $$\frac{dX_1}{dt} = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}, \quad \frac{dX_2}{dt} = \begin{pmatrix} -e^{-t} \\ -e^{-t} \end{pmatrix}$$

and $$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 3e^t \\ e^t \end{pmatrix} = \begin{pmatrix} 6e^t - 3e^t \\ 3e^t - 2e^t \end{pmatrix} = \begin{pmatrix} 3e^t \\ e^t \end{pmatrix} = \frac{dX_1}{dt}$$

Similarly

$$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix} = \begin{pmatrix} 2e^{-t} - 3e^{-t} \\ e^{-t} - 2e^{-t} \end{pmatrix} = \begin{pmatrix} -e^{-t} \\ -e^{-t} \end{pmatrix} = \frac{dX_2}{dt}$$

Hence both the vectors X_1 and X_2 are solutions of the homogeneous system

$$X' = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}X$$

Now suppose that c_1, c_2 are any two arbitrary real constants such that

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Very Important lecture

41 Matrices and Systems of Linear 1st-Order Equations (Continued)

41.1 Theorem

A necessary and sufficient condition that the set of solutions, on an interval I, consisting of the vectors

$$X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, X_2 = \begin{pmatrix} x_{12} \\ x_{21} \\ \vdots \\ x_{n2} \end{pmatrix}, \ldots, X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

Quiz

of the homogenous system $X' = AX$ to be linearly independent is that the Wronskian of these solutions is non-zero for every $t \in I$. Thus

$$W(X_1, X_2, \ldots, X_n) = \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{vmatrix} \neq 0, \quad \forall t \in I$$

Note that

- It can be shown that if $X_1, X_2, \ldots, X_n$ are solution vectors of the system, then either

  $$W(X_1, X_2, \ldots, X_n) \neq 0, \quad \forall t \in I$$

  Quiz

  or $$W(X_1, X_2, \ldots, X_n) = 0, \quad \forall t \in I$$

  Thus if we can show that $W \neq 0$ for some $t_0 \in I$, then $W \neq 0$, $\forall t \in I$ and hence the solutions are linearly independent on I.

  Quiz

- Unlike our previous definition of the Wronskian, the determinant does not involve any differentiation.

Example 1

Short

Important

As verified earlier that the vectors

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

are solutions of the following homogeneous system.

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

1. $\frac{dx}{dt} = x - y + z + t - 1$

   $\frac{dy}{dt} = 2x + y - z - 3t^2$

   $\frac{dz}{dt} = x + y + z + t^2 - t + 2$

2. $\frac{dx}{dt} = -3x + 4y + e^{-t}\sin 2t$

   $\frac{dx}{dt} = 5x + 9y + 4e^{-t}\cos 2t$

3. $\frac{dx}{dt} = -3x + 4y - 9z$

   $\frac{dy}{dt} = 6x - y$

   $\frac{dz}{dt} = 10x + 4y + 3z$

4. $\frac{dx}{dt} = -3x + 4y + e^{-t}\sin 2t$

   $\frac{dy}{dt} = 5x + 9y + 4e^{-t}\cos 2t$

Write the given system without of use of matrices

5. $X' = \begin{pmatrix} 7 & 5 & -9 \\ 4 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix} X + \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} e^{5t} - \begin{pmatrix} 8 \\ 0 \\ 3 \end{pmatrix} e^{-2t}$

6. $\frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix}\sin t + \begin{pmatrix} t-4 \\ 2t+1 \end{pmatrix} e^{4t}$

7. $\frac{d}{dt}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 3 & -4 & 1 \\ -2 & 5 & 6 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} e^{-t} - \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} t$

Verify that the vector X is the solution of the given system

8. $\frac{dx}{dt} = -2x + 5y$

   $\frac{dx}{dt} = -2x + 4y, \quad X = \begin{pmatrix} 5\cos t \\ 3\cos t - \sin t \end{pmatrix} e^t$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

Hence X_1 and X_2 form a fundamental set of solution of the system on the interval $(-\infty,\infty)$. Hence, the general solution of the system on $(-\infty,\infty)$ is

$$X = c_1X_1 + c_2X_2 = c_1\begin{pmatrix}1\\-2\end{pmatrix}e^{-2t} + c_2\begin{pmatrix}3\\5\end{pmatrix}e^{3t}$$

Example 3

Consider the vectors X_1, X_2 and X_3 these vectors are given by

$$X_1 = \begin{pmatrix}\cos t\\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t\\ -\cos t - \sin t\end{pmatrix},\ X_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}e^t,\ X_3 = \begin{pmatrix}\sin t\\ -\frac{1}{2}\sin t - \frac{1}{2}\cos t\\ -\sin t + \cos t\end{pmatrix}$$

It has been verified in the last lecture that the vectors X_1 and X_2 are solutions of the homogeneous system

$$X' = \begin{pmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ -2 & 0 & -1\end{pmatrix}X$$

It can be easily verified that the vector X_3 is also a solution of the system. We now compute the Wronskian of the solution vectors X_1, X_2 and X_3

$$W(X_1,X_2,X_3) = \begin{vmatrix}\cos t & 0 & \sin t\\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t & e^t & -\frac{1}{2}\sin t - \frac{1}{2}\cos t\\ -\cos t - \sin t & 0 & -\sin t + \cos t\end{vmatrix}$$

Expand from 2nd column

or
$$W(X_1,X_2,X_3) = e^t\begin{vmatrix}\cos t & \sin t\\ -\cos t - \sin t & -\sin t + \cos t\end{vmatrix}$$

or
$$W(X_1,X_2,X_3) = e^t \neq 0, \quad \forall t \in R$$

Thus, we conclude that X_1, X_2 and X_3 form a fundamental set of solution on $(-\infty,\infty)$. Hence, the general solution of the system on $(-\infty,\infty)$ is

$$X = c_1X_1 + c_2X_2 + c_3X_3$$

or

Scanned with CamScanner

Clearly, X_1 and X_2 are linearly independent on $(-\infty, \infty)$ as neither of the vectors is a constant multiple of the other. We now compute Wronskian of the solution vectors X_1 and X_2.

$$W(X_1, X_2) = \begin{vmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{vmatrix} = 8e^{4t} \neq 0, \quad \forall t \in (-\infty, \infty)$$

41.2 Fundamental set of solution

Suppose that $\{X_1, X_2, \ldots, X_n\}$ is a set of n solution vectors, on an interval I, of a homogenous system $X' = AX$. (The set is said to be a fundamental set of solutions of the system on the interval I if the solution vectors $X_1, X_2, \ldots, X_n$ are linearly independent.) Quiz

41.2.1 Theorem (Existence of a Fundamental Set)

There exist a fundamental set of solution for the homogenous system $X' = AX$ on an interval I

41.3 General solution

Quiz

Suppose that $X_1, X_2, \ldots, X_n$ is a fundamental set of solution of the homogenous system $X' = AX$ on an interval I. Then any linear combination of the solution vectors $X_1, X_2, \ldots, X_n$ of the form

$$X = c_1X_1 + c_2X_2 + \cdots + c_nX_n$$

$c_i; i = 1, 2, \ldots, n$ being arbitrary constants is said to be the general solution of the system on the interval I.

Quiz

Note that

For appropriate choices of the arbitrary constants $c_1, c_2, \ldots, c_n$ any solution, on the interval I, of the homogeneous system $X' = AX$ can be obtained from the general solution.

Important for short

Example 2

As discussed in the Example 1, the following vectors are linearly independent solutions

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}, \quad X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

of the following homogeneous system of differential equations on $(-\infty, \infty)$

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

or $$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} -12t+14 \\ -2 \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

or $$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \end{pmatrix} = X_p'$$

Thus the given vector X_p satisfies the non-homogeneous system of differential equations. Hence, the given vector X_p is a particular solution of the non-homogeneous system.

41.5 Theorem

Quiz

Let $X_1, X_2, \ldots, X_k$ be a set of solution vectors of the homogenous system $X' = AX$ on an interval I and let X_p be any solution vector of the non-homogenous system $X' = AX + F(t)$ on the same interval I. Then $\exists$ constants $c_1, c_2, \ldots, c_k$ such that

$$X_p = c_1X_1 + c_2X_2 + \ldots + c_kX_k + X_p$$

is also a solution of the non-homogenous system on the interval.

41.5.1 Complementary function

Let $X_1, X_2, \ldots, X_n$ be solution vectors of the homogenous system $X' = AX$ on an interval I, then the general solution

Short + Quiz

$$X = c_1X_1 + c_2X_2 + \ldots + c_nX_n$$

of the homogeneous system is called the complementary function of the non-homogeneous system $X' = AX + F(t)$ on the same interval I.

41.5.2 General solution of a Non homogenous systems

Let X_p be a particular integral and X_c the complementary function, on an interval I, of the non-homogenous system

Short + Quiz

$$X' = A(t)X + F(t).$$

The general solution of the non-homogenous system on the interval I is defined to be

$$X = X_c + X_p$$

Imp

Short or Long

Example 5

In Example 4 it was verified that

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$X = c_1\begin{pmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{pmatrix} + c_2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}e^t + c_3\begin{pmatrix} \sin t \\ -\frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\sin t + \cos t \end{pmatrix}$$

41.4 Non-homogeneous Systems

Short (2marks)

As stated earlier in this lecture that a system of differential equations such as

Quiz

$$\frac{dX}{dt} = A(t)X + F(t)$$

is non-homogeneous if $F(t) \neq 0, \ \forall t$. The general solution of such a system consists of a complementary function and a particular integral.

41.4.1 Particular Integral

Quiz

A particular solution, on an interval I, of a non-homogeneous system is any vector X_p free of arbitrary parameters, whose entries are functions that satisfy each equation of the system.

Importan For short and Long

Example 4

Show that the vector

$$X_p = \begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix}$$

is a particular solution of the following non-homogeneous system on the interval $(-\infty, \infty)$

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}X + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

Solution:

Differentiating the given vector with respect to t, we obtain

$$X_p' = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

Further

$$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}.$$

or

$$\begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}X_p + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix} = \begin{pmatrix} (3t-4)+3(-5t+6) \\ 5(3t-4)+3(-5t+6) \end{pmatrix} + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$X_p = \begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix}$$

is a particular solution, on $(-\infty, \infty)$, of the non-homogenous system

$$X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X + \begin{pmatrix} 12t-11 \\ -3 \end{pmatrix}$$

As we have seen earlier, the general solution of the associated homogeneous system i.e. the complementary function of the given non-homogeneous system is

$$X_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

Hence the general solution, on $(-\infty, \infty)$, of the non-homogeneous system is

$$X = X_c + X_p$$

$$X = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} + \begin{pmatrix} 3t-4 \\ -5t+6 \end{pmatrix}$$

Short or Quiz

41.6 Fundamental Matrix

Suppose that the a fundamental set of n solution vectors of a homogeneous system $X' = AX$, on an interval I, consists of the vectors

$$X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, X_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}, \ldots, X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

Then a fundamental matrix of the system on the interval I is given by

$$\phi(t) = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{pmatrix}$$

Short or Long

Example 6

As verified earlier, the following vectors

© Copyright Virtual University of Pakistan

Scanned with CamScanner

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$$

$$X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} = \begin{pmatrix} 3e^{6t} \\ 5e^{6t} \end{pmatrix}$$

form a fundamental set of solutions of the system on $(-\infty, \infty)$, $X' = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix} X$

So that the general solution of the system is $X = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$

Hence, a fundamental matrix of the system on the interval is $\phi(t) = \begin{pmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{pmatrix}$

Note that

- The general solution of the system can be written as Quiz

$$X = \begin{pmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \Rightarrow X = \phi(t)C, \quad C = (c_1 \quad c_2)^{tr}$$

- Since $X = \phi(t)C$ is a solution of the system $X' = A(t)X$. Therefore

$$\phi'(t)C = A(t)\phi(t)C \Rightarrow [\phi'(t) - A(t)\phi(t)]C = 0$$

Since the last equation is to hold for every t in the interval I for every possible co matrix of constants C, we must have $\phi'(t) - A(t)\phi(t) = 0 \Rightarrow \phi'(t) = A(t)\phi(t)$

Note that

- The fundamental matrix $\phi(t)$ of a homogenous system $X' = A(t)X$ is singular because the determinant $\det(\phi(t))$ coincides with the Wronskian o solution vectors of the system and linear independence of the solution ve guarantees that $\det(\phi(t)) \neq 0$.
- Let $\phi(t)$ be a fundamental matrix of the homogenous system $X' = A(t)X$ o interval I. Then, in view of the above mentioned observation, the inverse of matrix $\phi^{-1}(t)$ exists for every value of t in the interval I.

41.7 Exercise

The given vectors are the solutions of a system $X' = AX$. Determine whether the vec form a fundamental set on $(-\infty, \infty)$.

Scanned with CamScanner



Scanned with CamScanner